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APUL

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Classification of elements and Periodicity in Properties - Periodic Table

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1. Periodic Law is essentially the consequence of-
(1) Periodic similarities in electronic configuration
(2) Periodic variation in electronic configuration
(3) Periodic similarities in physical properties
(4) None of these

CPT343

2. Match the following

(A) 111	(i) Nihonium
(B) 113	(ii) Rontgenium
(C) 115	(iii) Tennessine
(D) 117	(iv) Moscovium

(1) A→(ii); B→(i); C→(iv); D→(iii)
(2) A→(i); B→(ii); C→(iv); D→(iii)
(3) A→(ii); B→(iv); C→(i); D→(iii)
(4) A→(ii); B→(i); C→(iii); D→(iv)

CPT344

3. **Assertion :-** There are eight elements in second period of periodic table.
Reason :- Generally number of elements in each period of Periodic table is Twice the number of atomic orbitals available in the energy level that is being filled.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CPT345

4. **Statement-1 :-**
Halogens and Chalcogens have highly negative electron gain enthalpy.
Statement-2 :-
Adding one or two electrons respectively they attain stable noble gas configuration.
(1) Statement I is correct but statement II is incorrect.
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and II are correct.
(4) Both statement I and II are incorrect.

CPT346

5. The elements z = 117 and z = 120 have not yet been discovered. In which group would you place these elements.
(1) z = 117 (Group 17)
(2) z = 120 (Group 2)
(3) Both (1) and (2)
(4) z = 120 (Group 13)

CPT347

6. Arrange the following elements in increasing order of metallic character, Si, Be, Mg, Na, P
(1) Si < Be < Mg < Na < P
(2) P < Si < Be < Mg < Na
(3) Be < Si < Mg < Na < P
(4) Be < Mg < Si < C < Na < P

CPT348

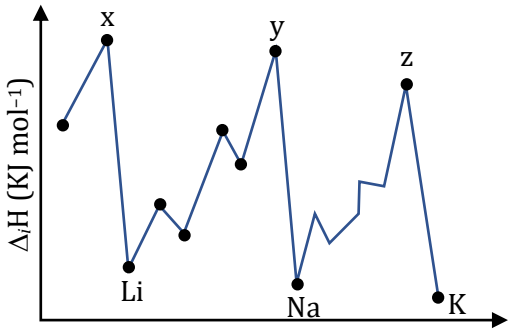
7. **Assertion :-** The atomic size generally decrease across a period.
Reason:- Within the period outer electrons are in same valence shell and Z_{eff} (effective nuclear charge) increases as the atomic no. increases, resulting in increased attraction of electrons to the nucleus.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CPT349

8. **Assertion :-** For alkali metals and Halogens atomic radius increases regularly with atomic number.
Reason :- As we descend the groups, the principal quantum number increases and the valence electrons are further from nucleus. This happens because inner energy levels are filled with electrons which serve to shield the outer electrons from pull of nucleus.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CPT350

9. Given below is the graph shown between $\Delta_i H$ (Ionisation enthalpy) and atomic number (z).



Identify the elements which are at the peaks of the curve.
(1) Alkali metals
(2) Alkaline earth metals
(3) Inert gases
(4) Halogens

CPT351

10. **Assertion :-** The first Ionisation Energy of Boron is slightly less than of Beryllium.
Reason :- The penetration of a 2s electron to the nucleus is more than that of a 2p electron Hence the 2p electron of Boron is more shielded from the nucleus by inner core of electrons i.e. 2s electron of Beryllium.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CPT352

11. **Assertion :-** First Ionisation Energy of Oxygen is smaller than Nitrogen.
Reason :- In Oxygen atom two of the four 2p electron must occupy the same 2p orbital resulting in an increased electron-electron repulsion.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CPT353

12. Arrange the following elements in decreasing order of their electron gain enthalpy -
O, S, Se, Te, Po
(1) S, Se, Te, Po, O (2) O, S, Se, Te, Po
(3) Po, O, Te, S, Se (4) Se, S, Te, Po, O

CPT354

13. Arrange the following inert gases in decreasing order of Electron Gain Enthalpy.
(1) He > Ne > Ar = Kr > Xe > Rn
(2) Ne > Ar = Kr > Xe > Rn > He
(3) Ne > Kr > Ar > Xe > He > Rn
(4) Rn > Ne > Kr > Ar > Xe > He

CPT355

14. **Assertion :-** Group-17 elements (the halogens) have very high negative electron gain enthalpies.
Reason :- Because they can attain stable noble gas electronic configurations by picking up an electron
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CPT356

15. **Assertion:-** Noble gases have large positive electron gain enthalpies.
Reason :- The electron has to enter the next higher principal quantum level leading to a very unstable electronic configuration.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CPT357

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16. **Assertion :-** Electron Gain Enthalpy of Oxygen or Fluorine is less negative than that of succeeding element.
Reason :- The added electron goes to smaller n = 2 quantum level and suffers significant repulsion from the other electron present in this level.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.
CPT358

17. Arrange the following in decreasing order of electron gain enthalpy
P, S, Cl, F
(1) F, Cl, S, P (2) P, S, Cl, F
(3) Cl, F, S, P (4) F, S, P, Cl
CPT359

18. Energy is released when an electron is added to an atom, the electron affinity is taken as.
(1) Zero
(2) Negative
(3) Positive
(4) can't be predicted
CPT360

19. **Statement-I :-** Electronegativity of a given element is not a measurable quantity.
Statement-II :- It does provide the means of predicting the nature of force that holds a pair of atoms together.
(1) Statement I is correct but statement II is incorrect.
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and II are correct.
(4) Both statement I and II are incorrect.
CPT361

20. The valency of representative elements is usually.
(1) Equal to number of electron in outermost orbitals.
(2) Equal to eight minus the number of outermost electrons.
(3) Both (1) and (2)
(4) None of these
CPT362

21. Lithium and Beryllium is more similar with Magnesium and Aluminium due to-
(1) Diagonal relationship
(2) Horizontal relationship
(3) Vertical relationship
(4) None of these
CPT363

22. Reasons for different chemical behaviour of the first member of a group of elements in the s and p block
(1) Small size
(2) Large charge to size ratio
(3) High electronegativity
(4) All of the above
CPT364

23. Size of ionic radius of Fluoride ion (F^-) is 136 pm then atomic radius of 'F' will be -
(1) 180 pm (2) 146 pm
(3) 64 pm (4) 250 pm
CPT365

24. **Assertion :-** Anion with greater negative charge will have larger radius.
Reason :- The net repulsion of e^- will out weigh the nuclear charge.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.
CPT366

25. Which of the following species will have largest and smallest size respectively Mg, Mg^{+2}, Al, Al^{+3}

(1) Mg, Al^{+3}

(2) Mg, Mg^{+2}

(3) Al, Al^{+3}

(4) Mg^{+2}, Al^{+3}

CPT367

26. Electronegativity is -

(1) Directly related to non-metallic properties.

(2) Inversely related to metallic properties.

(3) Both (1) and (2)

(4) No relation with metallic & non metallic properties.

CPT368

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ANSWER KEY

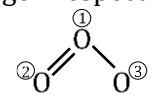
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**1**	**1**	**3**	**3**	**2**	**1**	**1**	**3**	**1**	**1**	**1**	**2**	**1**	**1**
Question	16	17	18	19	20	21	22	23	24	25	26				
Answer	**1**	**3**	**3**	**3**	**3**	**1**	**4**	**3**	**1**	**1**	**3**				

Chemical Bonding and Molecular Structure - Chemical Bonding

1. Which of the following statements does not agree with Lewis dot structure concept-
(1) Each bond is formed as a result of sharing of an e^- pair between the atoms.
(2) Each combining atom contributes atleast one e^- to the shared pair
(3) The combining atoms attains the outer most shell halogen configurations as a result of sharing of e^-
(4) All statements are correct

CCB520

2. Find correct set of formal charge on marked oxygen respectively.



(1) +1, +1, +1 (2) +1, –1, 0
(3) +1, 0, –1 (4) 0, +1, –1

CCB521

3. **Assertion :-** PF_5, SF_6, H_2SO_4 are considered as hypervalent molecules
Reason :- There are 10, 12, 12 electrons around P, S & S in PF_5, SF_6 and H_2SO_4 molecules respectively.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CCB522

4. Sulphur dichloride-
(1) Is a hyper valent molecule
(2) Is a hypo valent molecule
(3) Obeys octet rule
(4) None of the above

CCB523

5. Which of the following statement is correct about the octet theory.
(1) It is based upon the chemical inertness of noble gases with exceptions like XeF_2, KrF_2 etc.
(2) This theory does not account for the shape of molecules.
(3) It does not explain relative stability of the molecules.
(4) All of the above

CCB524

6. Ionic bond forms more easily between elements with comparatively.
(1) Low I.E. & high negative value of EGE
(2) High I.E. & high negative value of EGE
(3) High I.E. & low negative value of EGE
(4) Low I.E. & low negative value of EGE

CCB525

7. Identify the correct order of bond length in following
(1) O–H > C–H
(2) O–H < C–H
(3) O–H = C–H
(4) cannot be compared

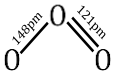
CCB526

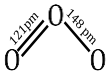
8. Compare bond length in following -
(1) C – N < C = N < C ≡ N
(2) C – N < C = N > C ≡ N
(3) C – N > C = N > C ≡ N
(4) C ≡ N < C – N < C = N

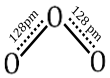
CCB527

9. Which of the following has identical bond orders ?
(1) F_2, O_2^{-2}
(2) CO, N_2
(3) NO^+, CO
(4) All of these

CCB528

10. Which structure represents structure of O_3 molecule more accurately.
(1) 

(2) 

(3) 

(4) All of the above

CCB529

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11. Which of the following statements is not correct ?
(1) The cannonical forms have no real existence.
(2) The molecule does not exist for a certain fraction of time in one cannonical form and for other fractions of time in other cannonical forms.
(3) There is an equilibrium between the cannonical forms as we have between tautomeric forms.
(4) The molecule as such has a single structure which is the resonance hybrid of the cannonical forms and which cannot as such depicted by a single Lewis structure.

CCB530

12. **Assertion :-** HCl being covalent molecule contain certain percentage of ionic character.
Reason :- In reality no bond or a compound is either completely covalent or ionic.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CCB531

13. In case of polyatomic molecule dipole moment depends on-
(1) Bond dipoles
(2) Spatial arrangement of various bonds in the molecules.
(3) Both A & B
(4) None of the above

CCB532

14. **Assertion :-** Dipole moment in case of BeF_2 is zero.
Reason :- Two equal bond dipoles point in opposite directions and cancel effect of each other.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CCB533

15. Identify the incorrect statements-
(1) Smaller the size of the cation and larger the size of the anion, greater the ionic character of the covalent bond.
(2) Greater the charge on the cation, greater the ionic character of the covalent bond.
(3) For cations of same size & charge, the one with electronic configuration (n–1) $d^n ns^0$ i.e. transition metals is more polarising than one with a noble gas configuration ns^2np^6 i.e. alkali and alkaline metal cations.
(4) Both (1) & (2)

CCB534

16. Match the column-

Column-I	**Column-II**
(A) AB	(i) CH_4
(B) AB_2	(ii) HF
(C) AB_3	(iii) NH_3
(D) AB_4	(iv) H_2S

(1) (A)→(ii); (B)→(iv); (C)→(iii); (D)→(i)
(2) (A)→(iv); (B)→(ii); (C)→(iii); (D)→(i)
(3) (A)→(ii); (B)→(iii); (C)→(iv); (D)→(i)
(4) (A)→(ii); (B)→(iv); (C)→(i); (D)→(iii)

CCB535

17. Compare the dipole moment of following molecules-
(1) $H_2O < H_2S < CO_2$
(2) HF < HCl < H-Br
(3) $NH_3 > NF_3 > BF_3$
(4) $CH_4 > CHCl_3 > CCl_4$

CCB536

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18. Match the column-

Column-I	**Column-II**
(A) H_2	(i) Trigonal-pyramidal
(B) NF_3	(ii) Linear
(C) $CHCl_3$	(iii) Bent
(D) H_2S	(iv) Tetrahedral

(1) (A)→(ii); (B)→(i); (C)→(iv); (D)→(iii)
(2) (A)→(i); (B)→(ii); (C)→(iv); (D)→(iii)
(3) (A)→(ii); (B)→(iv); (C)→(i); (D)→(iii)
(4) (A)→(ii); (B)→(i); (C)→(iii); (D)→(iv)

CCB537

19. Find incorrect statement about VSEPR theory ?

(i) The shape of molecule depends upon the no. of valence shell e^- pairs (bonded or non-bonded) around the central atom.

(ii) Pairs of e^- in the valence shell repel one another since their electron clouds are negatively charge.

(iii) These pairs of electrons tend to occupy such positions in shape that maximize repulsion.

(iv) VSEPR model is not applicable to any such structure where 2 or more resonance structure can represent a molecule.

(1) (i) & (ii)
(2) (ii) & (iii)
(3) (iii) & (iv)
(4) All are incorrect

CCB538

20. Incorrect order of bond angles-

(1) $BeCl_2 > BF_3$ (2) $BF_3 > CH_4$
(3) $CH_4 > BF_3$ (4) $CH_4 < BeCl_2$

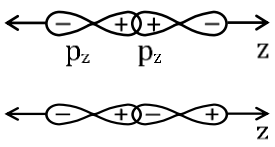
CCB539

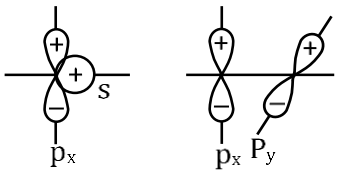
21. Find correct order of the following molecules in increasing order of lone pairs on central atom-

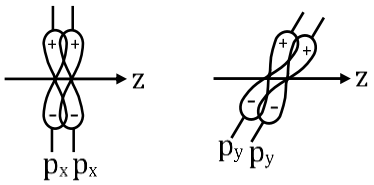
(1) $NH_3 < H_2O$ (2) $SF_4 < SO_2$
(3) $ClF_3 < BrF_5$ (4) $XeF_4 < NF_3$

CCB540

22. Which of the following set represents positive overlap.

(1) 

(2) 

(3) 

(4) All sets are correct

CCB541

23. Which among the following is not a salient features of hybridisation ?

(1) The number of hybrid orbital is equal to the number of atomic orbitals that gets hybridized.

(2) These hybrid orbitals are always equivalent in energy and shape.

(3) The hybrid orbitals are more effective in forming stable bond than pure atomic orbitals.

(4) The hybrid orbitals are directed in space to have maximum repulsion between electron pairs.

CCB542

24. Which of the following types of hybridization also termed as diagonal hybridization.

(1) sp
(2) sp^2
(3) sp^3d^2
(4) sp^3d

CCB543

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25. **Assertion :-** Axial bonds have been found to be slightly longer and hence slightly weaker the equatorial bonds-

Reason :- Axial bond pairs suffer more repulsive interaction from equatorial bonds.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

CCB544

26. **Assertion :-** PCl_5 molecules is more reactive.

Reason :- In PCl_5 axial bond suffer more repulsion interaction from equatorial bond pairs.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

CCB545

27. Which the following statements is/are incorrect regarding molecular orbital theory-

(A) The electrons are present in various atomic orbitals.

(B) The atomic orbitals are of comparable energies.

(C) e^- is influenced by 2 or more nuclei

(D) The bonding molecular orbital has higher energy than anti B.M.O.

(1) A & B (2) B & C

(3) C & D (4) A & D

CCB546

28. Which of the following relation represent a stable molecule in respect of MOT ?

(1) $N_b > N_a$ (2) $N_a < N_b$

(3) $N_b = N_a$ (4) None of these

CCB547

29. Correct order of bond order is -

(1) $H_2 > He_2 > Li_2$ (2) $He_2 > H_2 > C_2$

(3) $O_2 > Li_2 > He_2$ (4) $Li_2 > O_2 > C_2$

CCB548

30. Find the correct valence electronic configuration for the following molecule-

(1) B_2 :- $(\sigma_{2s})^2(\sigma^*_{2s})^2(\sigma_{2p})^2$

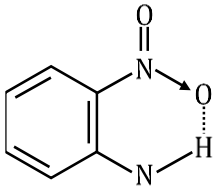
(2) O_2 :- $(\sigma_{2s})^2(\sigma^*_{2s})^2(\sigma_{2p})^4(\pi_{2p})^4(\pi^*_{2p})^2$

(3) N_2 :- $(\sigma_{2s})^1(\sigma^*_{2s})^1(\pi_{2p})^4(\sigma_{2p})^2$

(4) C_2 :- $(\sigma_{2s})^2(\sigma^*_{2s})^2(\pi_{2p})^4$

CCB549

31.



Which type of bond is represent in dotted line

(1) Coordinate bond

(2) Inter molecular H bonding

(3) Intra molecular H bonding

(4) Metallic bond

CCB550

32. The electrovalence is equal to

(1) Number of e^- of ion

(2) Number of p^+ of ion

(3) Number of n^0 of ion

(4) Number of unit charge an ion

CCB551

33. Among the following statements which is incorrect-

(1) The orbitals present in the valence shell of the atom are hybridized.

(2) The orbitals undergoing hybridization should have almost equal energy.

(3) Promotion of e^- is an essential condition prior to hybridization.

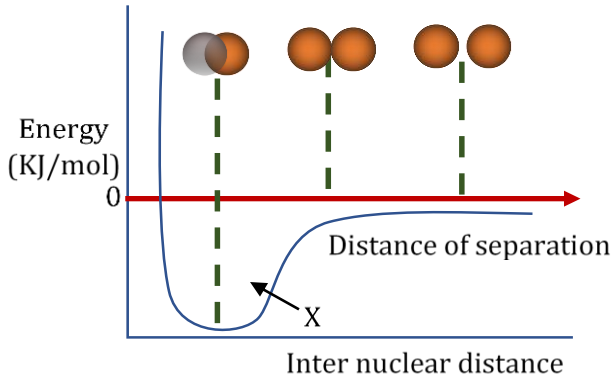
(4) Filled orbitals of valence shell can take part in hybridisation.

CCB552

34. **Assertion :-** The repulsion between the nuclei of an ABMO is high.
Reason :- There is nodal plane between the nuclei in ABMO
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CCB553

35.



Find 'x'
(1) Bond length
(2) Bond order
(3) Bond energy
(4) Bond disorder

CCB554

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**3**	**1**	**3**	**4**	**1**	**2**	**3**	**4**	**3**	**3**	**1**	**3**	**1**	**4**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**1**	**3**	**1**	**3**	**3**	**1**	**3**	**4**	**1**	**1**	**1**	**4**	**1**	**3**	**4**
Question	31	32	33	34	35										
Answer	**3**	**4**	**3**	**1**	**3**										

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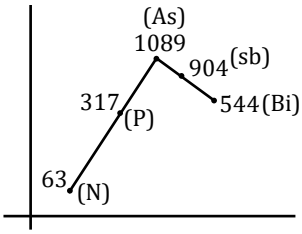
p-Block Elements - Group 13 to 18

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1. **Assertion :-** In Nitrogen family metallic character increases down the group.
Reason :- Due to decrease in I.P. and increase in Atomic size.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CPB437

2.



Above pattern shows the variation in which of the following-
(1) Boiling point (2) Melting point
(3) Ionisation enthalpy (4) Covalent radius

CPB438

3. Which of the following element does not show allotropy ?
(1) Nitrogen (2) Phosphorous
(3) Arsenic (4) Antimony

CPB439

4. **Assertion :-** The tendency to exhibit '-3' oxidation state decreases down the group in group 15.
Reason :- Size and metallic character increase down the group in group 15.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CPB440

5. In which of the following reaction Nitrogen disproportionate.
(1) $3HNO_2 \rightarrow HNO_3 + H_2O + 2NO$
(2) $N_2 + 3N_2 \rightarrow 2NH_3$
(3) $2NO_2 \underset{\text{Heat}}{\overset{\text{Cool}}{\rightleftharpoons}} N_2O_4$
(4) $4HNO_3 + P_4O_{10} \rightarrow 4HPO_3 + 2N_2O_5$

CPB441

6. Identify the correct order of reducing character for hydrides-
(1) $NH_3 < PH_3 > AsH_3 > BiH_3$
(2) $NH_3 > PH_3 < AsH_3 > BiH_3$
(3) $NH_3 < PH_3 < AsH_3 < BiH_3$
(4) $NH_3 > PH_3 > AsH_3 > BiH_3$

CPB442

7. Order of boiling point for the hydrides will be-
(1) $NH_3 > SbH_3 > AsH_3 > PH_3$
(2) $SbH_3 > NH_3 > AsH_3 > PH_3$
(3) $NH_3 > PH_3 > SbH_3 > AsH_3$
(4) $NH_3 > PH_3 > AsH_3 > SbH_3$

CPB443

8. The acidic order of the oxides of the Nitrogen family going down the group.
(1) Increase
(2) Decrease
(3) Remains constant
(4) None of these

CPB444

9. Catalyst used in manufacture of Ammonia-
(1) Iron oxide with small amount of K_2O & Al_2O_3
(2) Zinc oxide with small amount of K_2O & Al_2O_3
(3) Copper oxide with small amount of K_2O & Al_2O_3
(4) Sodium peroxide with small amount of K_2O & Al_2O_3

CPB445

10. Correct statement regarding HNO_3 is -
(1) It has octahedral geometry
(2) Its hybridization is sp^3d
(3) It exists as planar molecule
(4) It has $p\pi$-$d\pi$ multiple bond

CPB446

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11. For the given reactions
$3Cu + 8HNO_3(Dilute) \rightarrow 2Cu(NO_3)_2 + X + H_2O$
$Cu + 4HNO_3(Conc.) \rightarrow Cu(NO_3)_2 + Y + 2H_2O$
Gases X & Y are being produced.
Identify the nature of gases
(1) Paramagnetic, Diamagnetic
(2) Diamagnetic, Paramagnetic
(3) Paramagnetic, Paramagnetic
(4) Diamagnetic, Diamagnetic

CPB447

12. For the reaction.
$[Fe(H_2O)_6]^{+2} + NO \rightarrow [Fe(H_2O)_5(NO)]^{+2} + H_2O$
(X)
Identify the colour of 'X'
(1) Black (2) Green
(3) Brown (4) Purple

CPB448

13. Allotrope of phosphorous, which shows chemiluminescence-
(1) White phosphorus
(2) Red phosphorus
(3) Black phosphorus
(4) Blue phosphorus

CPB449

14. Which of the following does not holds true for phosphine-
(1) Its formula is PH_4^+
(2) It is a colourless gas
(3) It's smell is like a rotten fish
(4) It is used in Holme's signal & smoke screens.

CPB450

15. In solid state what is the shape of anionic part of PCl_5
(1) Tetrahedral (2) Square planar
(3) Hexagonal (4) Octahedral

CPB451

16. In the reaction
$4AgNO_3 + 2H_2O + H_3PO_2 \rightarrow 4Ag + 4HNO_3 + H_3PO_4$
Hypo-phosphorous acid acts as-
(1) Reducing agent (2) Oxidising agent
(3) Catalyst (4) All of these

CPB452

17. The Basicity of H_3PO_4 is –
(1) 1 (2) 2
(3) 3 (4) 4

CPB453

18. Name for group 16 (oxygen family) is -
(1) Pnicogens (2) Chalcogens
(3) Halogens (4) Heavygens

CPB454

19. **Assertion :-** Elements of group 16 generally show lower value of first ionisation enthalpy compared to the corresponding periods of group 15.
Reason : - Due to extra stable half-filled p orbitals electronic configurations of Group 15 elements, larger amount of energy is required to remove electrons compared to Group 16 elements.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CPB455

20. Identify the correct increasing order of acidic character-
(1) $H_2O > H_2S > H_2Se > H_2Te$
(2) $H_2O > H_2S < H_2Se > H_2Te$
(3) $H_2O < H_2S < H_2Se < H_2Te$
(4) $H_2O < H_2S > H_2Se > H_2Te$

CPB456

21. Find correct increasing order of reducing property-
(1) $H_2S < H_2Se < H_2Te$
(2) $H_2S > H_2Se > H_2Te$
(3) $H_2S < H_2Se > H_2Te$
(4) None of these

CPB457

22. Correct order of increasing melting point-
(1) $H_2O > H_2Te > H_2Se > H_2S$
(2) $H_2O < H_2Te > H_2Se < H_2S$
(3) $H_2O < H_2Te < H_2Se < H_2S$
(4) None of these

CPB458

23. The magnetic nature of dioxygen is -
(1) Diamagnetic
(2) Paramagnetic
(3) Both A & B
(4) None of these

CPB459

24. In the following reaction identify the role of O_2

$2SO_2 + O_2 \xrightarrow{V_2O_5} 2SO_3$

$4HCl + O_2 \xrightarrow{CuCl_2} 2Cl_2 + 2H_2O$

(1) As a catalyst
(2) As a reducing agent
(3) As an oxidising agent
(4) Bleaching agent

CPB460

25. For the reaction

$3O_2 \rightarrow 2O_3$ T = 298 K

Identify the nature of reaction.
(1) Exothermic (2) Endothermic
(3) ΔH negative (4) None of these

CPB461

26. Form of sulphur which shows paramagnetic behaviour-
(1) S_6 (2) S_8
(3) S_2 (4) S_{10}

CPB462

27. Identify the catalyst in the following reaction.

$2SO_{2(g)} + O_{2(g)} \longrightarrow 2SO_{3(g)}$

(1) Charcoal (2) V_2O_5
(3) CrO_2Cl_2 (4) $FeCl_3$

CPB463

28. In the reaction charring of sugar, H_2SO_4 act as-
(1) Catalyst
(2) Moisturiser
(3) Dehydrating agent
(4) All of these

CPB464

29. Halogens have maximum _____ electron gain enthalpy in the respective periods of the periodic table.
(1) Positive (2) Negative
(3) Can not be said (4) Contain any value

CPB465

30. Aqua regia is used to dissolve :-
(1) Noble gases
(2) Transition metals
(3) Noble metals
(4) Noble non-metals

CPB466

31. Hydrolysis of XeF_6 is -
(1) A non-redox reaction
(2) A combustion reaction
(3) A redox reaction
(4) All of the above

CPB467

32. Match the Column-

Compounds		Formula	
(A)	Orthoboric acid	(P)	$Na_2B_4O_7.4H_2O$
(B)	Borax	(Q)	B_2H_6
(C)	Kernite	(R)	H_3BO_3
(D)	Diborane	(S)	$Na_2B_4O_7.10H_2O$

(1) A → R; B→S; C→P; D→Q
(2) A → S; B→R; C→P; D→Q
(3) A → R; B→P; C→S; D→Q
(4) A → R; B→S; C→Q; D→P

CPB468

33. **Assertion :-** Atomic radius of Gallium is less than that of Aluminium.
Reason :- Presence of additional 'd' electrons in gallium offers only poor screening effect for the outer electron.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CPB469

34. Correct order of I.E. is-
(1) B > Ga > Al > In > Tl
(2) B > Tl > Ga > Al > In
(3) B < Tl < Ga < Al < In
(4) B < Ga < Al < In < Tl

CPB470

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35. Correct order of Melting point-
(1) B < Al < Tl < In < Ga
(2) B < Tl < Al < In < Ga
(3) B > Al > Tl > In > Ga
(4) B > Tl > Al > Ga > In

CPB471

36. Trichlorides of Boron on hydrolysis in water form-
(1) $[M(OH)_4]^-$ (2) $[M(OH)_3]^-$
(3) $[M(OH)_4]^+$ (4) $[M(OH)_3]^+$

CPB472

37. In the reaction
$2MH + B_2H_6 \rightarrow 2M+[BH_4]^-$
Identify 'M'-
(1) Ra, Fr (2) Li, Be
(3) Li, Na (4) Na, Mg

CPB473

38. **Assertion :-** Compounds in '+4' oxidation state of carbon are generally covalent in nature.
Reason :- Sum of first four I.P. is very high.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CPB474

39. Match the column-

Species		Formula	
(A)	Water gas	(P)	$CO + N_2$
(B)	Producer gas	(Q)	$CO + H_2$
(C)	Natural gas	(R)	CO_2
(D)	Dry ice	(S)	CH_4

(1) A → P; B → Q; C → R; D → S
(2) A → Q; B → P; C → S; D → R
(3) A → P; B → R; C → Q; D → S
(4) A → P; B → Q; C → S; D → R

CPB475

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**1**	**2**	**1**	**1**	**1**	**3**	**2**	**2**	**1**	**3**	**3**	**3**	**1**	**1**	**4**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**1**	**3**	**2**	**1**	**3**	**1**	**1**	**2**	**3**	**2**	**3**	**2**	**3**	**2**	**3**
Question	31	32	33	34	35	36	37	38	39						
Answer	**1**	**1**	**1**	**2**	**3**	**1**	**3**	**1**	**2**						

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d and f Block Elements

1. The outermost electronic configuration of 'Pd' is-
(1) $4d^9 5s^1$ (2) $4d^{10} 5s^0$
(3) $4d^8 5s^2$ (4) None of these
CDF164

2. **Assertion :-** Silver is a transition element.
Reason :- Silver atom has completely filled 'd' orbitals ($4d^{10}$) in its ground state.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.
CDF165

3. Which of the following point does not hold true for any transition metals ?
(1) Very hard
(2) Low volatility
(3) High melting point and boiling point
(4) Contains fully filled 'd' orbitals
CDF166

4. **Assertion :-** Melting point of the transition elements are generally high.
Reason :- Involvement of greater number of electrons from (n-1)d in addition to the 'ns' electrons in the interatomic metallic bonding.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.
CDF167

5. Which of the following statement is incorrect in context of Enthalpy of atomisation.
(1) One unpaired electron per 'd' orbitals is particularly favourable for strong inter atomic interaction.
(2) Metals of the 2nd and 3rd series of d-Block have greater enthalpies of atomisation than the corresponding elements of the first series.
(3) Enthalpy of atomisation is an important factor in determining the standard electrode potential.
(4) Lesser number of valence electrons, stronger is the resultant bonding.
CDF168

6. Correct order of atomic radius will be-
(1) Zr < Hf
(2) Zr = Hf
(3) Zr > Hf
(4) Cann't be compared.
CDF169

7. Which term is not responsible for the value of ionisation enthalpy of 'd' block elements-
(1) Hybridization
(2) Attraction of each electron towards nucleus.
(3) Repulsion between the electrons.
(4) Exchange energy
CDF170

8. True statement about variability in oxidation states of transition elements.
(1) Their oxidation states differ from each other by unit two.
(2) Their oxidation states differ from each other by unit three.
(3) Their oxidation states differ from each other by unit four.
(4) Their oxidation states differ from each other by unit one.
CDF171

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9. MoO_3 and WO_3 will act not as _____ as like Cr(VI) in form of dichromate in acidic medium.
(1) Strong oxidizing agent
(2) Strong reducing agent
(3) Both 1 & 2
(4) None of the above

CDF172

10. Which of the following oxidising acids are being reduced on reaction with copper.
(1) Nitric acid
(2) Hot concentrated sulphuric acid
(3) Both (1) and (2)
(4) Dilute sulphuric acid

CDF173

11. **Assertion :-** Cr^{+2} is reducing in nature having d^4 configuration.
Reason :- Configuration of Cr^{+2} changes from d^4 to d^3 achieving half-filled t_{2g} level.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CDF174

12. The $E°(M^{2+}/M)$ value for copper is positive (+0.340)
The possible reason is -
(1) Hydration energy of Cu^{+2} is not enough to compensate sublimation energy, IE_1 and IE_2 for Cu.
(2) Lattice energy of Cu is not enough to compensate sublimation energy, IE_1 and IE_2 for Cu.
(3) Both (1) and (2)
(4) None of the above

CDF175

13. Which of the following compound formation is possible -
(1) MnO_3F (2) FeX_3
(3) CoF_3 (4) All of the above

CDF176

14. **Assertion :-** The fluorine has ability to stabilise the highest oxidation state as in the case of CoF_3.
Reason :- Due to higher lattice energy.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CDF177

15. All Cu(II) halides are known except -
(1) Bromide (2) Iodide
(3) Both (1) and (2) (4) None of these

CDF178

16. Find the correct order of oxidizing power ?
(1) $VO_2^+ < Cr_2O_7^{-2} < MnO_4^-$
(2) $VO_2^+ > Cr_2O_7^{-2} > MnO_4^-$
(3) $Cr_2O_7^{-2} > VO_2^+ > MnO_4^-$
(4) $MnO_4^- > VO_2^+ > Cr_2O_7^{-2}$

CDF179

17. The spin only magnetic moment of $M^{+2}_{(aq)}$ ion (z = 27) is -
(1) $\sqrt{25}$ (2) $\sqrt{15}$
(3) $\sqrt{36}$ (4) $\sqrt{17}$

CDF180

18. **Assertion :-** The transition metals and their compounds are known for their catalytic activity.
Reason :- Transition metals adopt multiple oxidation states and form complexes.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CDF181

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19. Match the column-

Catalyst	**Process**
(A) Vanadium (V) oxide	(P) Haber's process
(B) Finely divided iron	(Q) Contact process
(C) Nickel	(R) Hydrogenation
(D) $CuCl_2$	(S) Deacon's process

(1) A → Q; B → P; C → R; D → S
(2) A → P; B → Q; C → R; D → S
(3) A → Q; B → R; C → P; D → S
(4) A → Q; B → P; C → S; D → R

CDF182

20. **Statement-I :-** All metals except scandium forms 'MO' oxides
Statement-II :- MO oxides of transition metals are ionic in nature.
(1) Statement I is correct but statement II is incorrect.
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and II are correct.
(4) Both statement I and II are incorrect.

CDF183

21. Match the column –

	Species		**Uses**
(A)	Ferrous alloy	(P)	Leather industry in preparation of azo compound
(B)	$K_2Cr_2O_7$	(Q)	To produce bullets, shells and lighter Flint
(C)	Misch metal	(R)	Dry battery cell
(D)	MnO_2	(S)	Production of variety of steels and stainless steels

(1) A → S; B → P; C → Q; D → R
(2) A → P; B → S; C → Q; D → R
(3) A → S; B → Q; C → P; D → R
(4) A → S; B → P; C → R; D → Q

CDF184

22. Consider the following reaction

$$4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow \underset{(X)}{8Na_2CrO_4} + 2Fe_2O_3 + 8CO_2$$

Identify the colour of 'X'
(1) Red (2) Apple Green
(3) Indigo (4) Yellow

CDF185

23. Identify the shapes of chromate and dichromate ions-
(1) Trigonal planar and 2 Trigonal planar units sharing on corner with Cr–O–Cr
(2) Tetrahedral and 2 tetrahedral units sharing on corner with Cr–O–Cr
(3) Pentagonal bipyramidal 2 Tetrahedral planar units sharing on corner with Cr–O–Cr
(4) Structure not possible

CDF186

24. Which compounds is preferred over $Na_2Cr_2O_7$ as a primary standard in volumetric analysis.
(1) Na_2CrO_4
(2) K_2CrO_4
(3) $K_2Cr_2O_7$
(4) All of the above can be used.

CDF187

25. Identify the commercial method of preparation of MnO_4^-
(1) Electrolytic reduction in acidic solution
(2) Electrolytic oxidation in acidic solution
(3) Electrolytic reduction in alkaline solution
(4) Electrolytic oxidation in alkaline solution

CDF188

26. Match the column-

Compounds	**Colour**
(A) $K_2Cr_2O_7$	(P) Yellow
(B) $KMnO_4$	(Q) Orange
(C) Sodium chromate	(R) Dark purple
(D) Manganate	(S) Green

(1) A → Q; B → R; C → P; D → S
(2) A → R; B → Q; C → P; D → S
(3) A → Q; B → P; C → R; D → S
(4) A → Q; B → R; C → S; D → P

CDF189

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27. "Permanganate titrations in presence of HCl are unsatisfactory" identify the reason.
(1) HCl is oxidised to chlorine
(2) HCl is reduced to chlorine
(3) Titration not possible
(4) Both (1) and (2)
CDF190

28. Which of the following is/are not the use of potassium permanganate-
(1) Bleaching of wool
(2) Decolourisation of oils
(3) Colouring of Silk
(4) As a oxidant
CDF191

29. Oxidation states predominant in lanthanum of lanthanoids are-
(1) La (III) and Ln (III) (2) La (III) and Ln (II)
(3) La (II) and Ln (III) (4) La (0) and Ln (II)
CDF192

30. Match the column-

Compounds		Uses	
(A)	TiO	(P)	Oxidation of ethene to ethanal
(B)	V_2O_5	(Q)	Pigment industry
(C)	Zeigler Natta catalysts	(R)	Catalysis the oxidation of SO_2
(D)	$PdCl_2$ (Wacker process)	(S)	Polythene manufacture

(1) A → Q; C → S; B → R; D → P
(2) A → S; C → Q; B → R; D → P
(3) A → Q; C → R; B → S; D → P
(4) A → Q; C → P; B → S; D → R
CDF193

31. **Assertion :-** In general ions of same charge in a given 'd' series show progressive decrease in radius with increasing atomic number.
Reason :- The new electron enters a 'd' orbital each time the nuclear charge increases by unity.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.
CDF194

32. **Statement-I :-** Transition metals are defined as metals which have incomplete "d subshell" either in neutral atom or in the ions.
Statement-II:- Zinc, Cadmium and Mercury of group 12 are not regarded as transition metals.
(1) Statement I is correct but statement II is incorrect.
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and II are correct.
(4) Both statement I and II are incorrect.
CDF195

33. Ionisation becomes more different when-
(1) There is loss of exchange energy.
(2) There is gain of exchange energy.
(3) Stability is increase.
(4) (1) and (3) Both **CDF196**

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**2**	**4**	**1**	**4**	**3**	**1**	**4**	**1**	**3**	**1**	**1**	**4**	**1**	**2**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**1**	**2**	**1**	**1**	**3**	**1**	**4**	**2**	**3**	**4**	**1**	**1**	**3**	**3**	**1**
Question	31	32	33												
Answer	**1**	**3**	**4**												

Coordination Compounds

1. Match the column-

Compounds		Electrolyte	
(A)	$CoCl_3.6NH_3$	(P)	1:2 electrolyte
(B)	$CoCl_3.5NH_3$	(Q)	1:3 electrolyte
(C)	$CoCl_3.4NH_3$	(R)	1:1 electrolyte
(D)	$CoCl_3.3NH_3$	(S)	No electrolyte

(1) A→Q; B→P; C→R; D→S
(2) A→P; B→Q; C→R; D→S
(3) A→Q; B→R; C→P; D→S
(4) A→Q; B→P; C→S; D→R

CCO356

2. Which of the following statement is not considered as a postulate of Werner's theory-
(1) In coordination compounds metals show two types of linkages (valencies)-primary and secondary.
(2) The primary valency are normally ionisable and satisfied by negative ions.
(3) The secondary valency are non ionisable. These are satisfied by neutral molecules or negative ions. The secondary valence is equal to the coordination number and is fixed for a metal.
(4) The ions/groups bound by the primary and secondary linkages to the metal have characteristic spatial arrangements corresponding to different coordination numbers.

CCO357

3. How many compounds from following represents a double salt ?
(i) Carnallite (ii) Mohr's salt (iii) Potash Alum (iv) $K_4[Fe(CN)_6]$
(1) 1 (2) 2 (3) 3 (4) 4

CCO358

4. Find the incorrect relation
(1) SCN^- → Ambidentate ligand
(2) $N(CH_2CH_2NH_2)_3$ → polydentate ligand
(3) H_2O → Uni dentate ligand
(4) $EDTA^{4-}$ → didentate ligand

CCO359

5. Find the coordination number and oxidation state of 'Fe' for the complex $[Fe(C_2O_4)_3]^{-3}$
(1) 6, +3 (2) 5, +2 (3) 4, +3 (4) 3, +6

CCO360

6. The correct IUPAC nomenclature for the compound $[NiCl_2(PPh_3)_2]$ will be-
(1) di-chloridodi (tri-phenyl phosphine) nickel (II)
(2) di-chlroidodi (di-phenyl phosphine) nickel (II)
(3) di-chloridobis (tri-phenyl phosphine) nickel (II)
(4) di-chlorido Tris (triphenyl phosphine) nickel (II)

CCO361

7. Correct IUPAC name for the compound $[Co(H_2NCH_2CH_2NH_2)_3]_2(SO_4)_3$
(1) tris (ethane-1-2-diamine) cobalt (III) sulphate.
(2) Quad (ethane-1-2-diamine) cobalt (III) sulphate.
(3) tris (ethane-1-3-diamine) cobalt (III) sulphate.
(4) Quad (ethane-1-3-diamine) cobalt (III) sulphate.

CCO362

8. Match the column IUPAC

(A)	Tetraamminediaquacobalt(III) chloride	(P)	$[Cr(en)_3]Cl_3$
(B)	Amminebromidochloridonitrito-N-platinate (II)	(Q)	$[Pt(NH_3)BrCl(NO_2)]^-$
(C)	Tris(ethane–1,2–diamine) chromium (III) chloride	(R)	$[Co(H_2O)_2(NH_3)_4]Cl_3$
(D)	Dichloridobis(ethane–1,2–diamine) platinum (IV) nitrate	(S)	$[PtCl_2(en)_2](NO_3)_2$

(1) A→R; B→Q; C→P; D→S
(2) A→Q; B→R; C→P; D→S
(3) A→R; B→P; C→Q; D→S
(4) A→R; B→Q; C→S; D→P

CCO363

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9. Number of Geometrical isomers formed by the compound $[Co(NH_3)_4Cl_2]^+$ will be.
(1) No GI (2) 3 (3) 2 (4) 4

CCO364

10. Which of the following molecule cannot exhibit geometrical isomerism.
(1) $[CoCl_2(en)_2]$
(2) $[Co(NH_3)_3(NO_2)_3]$
(3) $[Cu(en)_2]^{+2}$
(4) $[Pt(NH_3)_2Cl_2]$

CCO365

11. Which of the following coordination entities not chiral (optically inactive)
(1) Cis-$[CrCl_2(ox)_2]^{-3}$
(2) trans-$[CrCl_2(ox)_2]^{-3}$
(3) $[PtCl_2(en)_2]^{+2}$
(4) $[Co(en)_3]^{+3}$

CCO366

12. Match the column-

Compound		Isomerism	
(A)	$[Co(NH_3)_5(NO_2)]Cl_2$	(P)	Solvate
(B)	$[Co(NH_3)_6][Cr(CN)_6]$	(Q)	Ionisation
(C)	$[Co(NH_3)_5SO_4]Br$ and $[Co(NH_3)_5Br]SO_4$	(R)	Linkage
(D)	$[Cr(H_2O)_6]Cl_3$	(S)	Coordination

(1) A→R; B→S; C→Q; D→P
(2) A→S; B→R; C→Q; D→P
(3) A→R; B→Q; C→S; D→P
(4) A→R; B→S; C→P; D→Q

CCO367

13. Predict the incorrect statement about $[Co(NH_3)_6]^{+3}$
(1) It is low spin complex
(2) Its hybridization is sp^3d^2
(3) It is diamagnetic in nature
(4) Its geometry is octahedral

CCO368

14. Match the column-

Compound		Property	
(A)	$[CoF_6]^{-3}$	(P)	d^2sp^3
(B)	$[NiCl_4]^{-2}$	(Q)	dsp^2
(C)	$[Co(NH_3)_6]^{+3}$	(R)	sp^3d^2
(D)	$[Ni(CN)_4]^{-2}$	(S)	sp^3

(1) A → R; B → S; C → P; D → Q
(2) A → S; B → R; C → P; D → Q
(3) A → R; B → P; C → S; D → Q
(4) A → R; B → S; C → Q; D → P

CCO369

15. Find correct statement about $[MnBr_4]^{-2}$
(1) Its shape is square planar
(2) It's hybridization is dsp^2
(3) Its magnetic moment is 5.9 B.M.
(4) Coordination number of Mn^{+2} in the complex is 6

CCO370

16. Which of the following is not a limitation of valence bond theory ?
(1) It does not explain the colour exhibited by coordination compounds.
(2) It does not make exact predictions regarding the tetrahedral and square planar structures of 4-coordinate complexes.
(3) It does not distinguish between weak and strong ligands.
(4) Metal atom or ion under influence of ligands can use orbitals for hybridisation.

CCO371

17. Which of the following is the correct order of increasing ligand field strength to form coordination compounds ?
(1) $CN^- < C_2O_4^{-2} < SCN^- < F^-$
(2) $SCN^- < F^- < C_2O_4^{-2} < CN^-$
(3) $SCN^- < F^- < CN^- < C_2O_4^{-2}$
(4) $F^- < SCN^- < C_2O_4^{-2} < CN^-$

CCO372

18. The crystal field stabilisation energy (CFSE) for $[CoCl_6]^{-4}$ is 18000 cm^{-1}. The CFSE for $[CoCl_4]^{-2}$ will be-
(1) 6000 cm^{-1} (2) 18000 cm^{-1}
(3) 16000 cm^{-1} (4) 8000 cm^{-1}

CCO373

19. What is the correct electronic configuration of the central atom in $K_4[Fe(CN)_6]$ based on crystal field theory.
(1) $t_{2g}^4eg^2$ (2) $t_{2g}^6eg^0$ (3) $e^3t_2^3$ (4) $e^4t_2^2$

CCO374

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20. Arrange the following in decreasing order of the absorbed wavelength ?
(1) $[Co(NH_3)_6]^{+3} > [Co(CN)_6]^{-3} > [Cu(H_2O)_4]^{+2}$
(2) $[Cu(H_2O)_4]^{+2} > [Co(CN)_6]^{-3} > [Co(NH_3)_6]^{+3}$
(3) $[Cu(H_2O)_4]^{+2} > [Co(NH_3)_6]^{+3} > [Co(CN)_6]^{-3}$
(4) All will have equal wavelength

CCO375

21. Compound $[Co(NH_3)_5(H_2O)]^{+3}$ absorb blue green color than its complementary color will be-
(1) Red (2) Voilet
(3) Yellow (4) Purple

CCO376

22. The number of unpaired electron of 'Pt' ion and the geometry of the complex $[Pt(CN)_4]^{-2}$.
(1) 0, octahedral
(2) 2, tetrahedral
(3) 0, square planar
(4) 2, pentagonal bipyramidal

CCO377

23. Decacarbonyldimanganese(0) is made up of -
(1) 2 Tetrahedral $Mn(CO)_5$ units.
(2) 2 Square pyramidal $Mn(CO)_5$ units.
(3) 2 Sqaure planar $Mn(CO)_5$ units.
(4) 2 Pentagonal bipyramidal $Mn(CO)_5$ units.

CCO378

24. Which of the following carbonyls will have the strongest C-O bond ?
(1) $[Fe(CO)_5]$ (2) $[Mn(CO)_6]^+$
(3) $[Cr(CO)_6]$ (4) $[V(CO)_6]^-$

CCO379

25. Hardness of water is estimated simple titration with-
(1) Na[EDTA] (2) Na_2[EDTA]
(3) Na_3[EDTA] (4) Na_4[EDTA]

CCO380

26. Which of the following is a coordination compound of cobalt ?
(1) Vitamin B_{12}
(2) Chlorophyll
(3) Anti-pernicious anaemia factor
(4) Both (1) and (3)

CCO381

27. $(Ph_3P)_3$ RhCl used as Wilkinson catalyst in-
(1) Deacons process
(2) Contact process
(3) Ostwald's process
(4) Hydrogenation of alkenes

CCO382

28. The complex formed in black and white photography is -
(1) $[(AgS_2O_3)_2]^{-3}$
(2) $[Au(S_2O_3)_2]^{-3}$
(3) $[Pt(S_2O_3)_2]^{-3}$
(4) $[Hg(S_2O_3)_2]^{-3}$

CCO383

29. Excess of copper and iron are removed by chelating ligand(s)____
(1) D–penicillamine
(2) Desferrioxamine B
(3) Chlorophyll
(4) Both A & B

CCO384

30. Compound used in the treatment of lead poisoning?
(1) Cis platin
(2) EDTA
(3) D-Penicillamine
(4) Trans platin

CCO385

31. **Assertion (A) :-** Emerald is gemstone, transmit light in green region.
Reason (R) :- In emerald, Cr^{+3} ions ions occupy octahedral sites in the mineral beryl ($Be_3Al_2Si_6O_{18}$).
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CCO386

32. Which of the following is not a limitation of the crystal field theory ?

(1) It assumes that ligands are point charges.

(2) It does not take into account the covalent character of bonding between ligand and the central atom.

(3) Formation of structures, colour and magnetic properties can be explained by the crystal field theory.

(4) All of the above are limitations of crystal field theory.

CCO387

33. Compound $[Co(NH_3)_5(H_2O)]^{+3}$ absorb blue green color than its complementary color will be-

(1) Red (2) Voilet

(3) Yellow (4) Purple

CCO388

34. In how many compounds that are given below the moles of AgCl precipitated per mole the compounds with excess $AgNO_3$ will be

(i) $PdCl_2.4NH_3$, (ii) $NiCl_2.6H_2O$,

(iii) Potash Alum, (iv) $K_4[Fe(CN)_6]$

(1) 2 compounds (2) 3 compounds

(3) No such compound (4) 1 compound

CCO389

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**1**	**4**	**3**	**4**	**1**	**3**	**1**	**1**	**3**	**3**	**2**	**1**	**2**	**1**	**3**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**4**	**2**	**4**	**2**	**3**	**1**	**3**	**2**	**2**	**2**	**4**	**4**	**1**	**4**	**2**
Question	31	32	33	34											
Answer	**1**	**3**	**1**	**1**											

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